Manel MERABET
Toulouse, France | +33 7 51 29 95 80 | manel.merabet@etu.toulouse-inp.fr | LinkedIn
First-year Electronics, Electrical Engineering, Energy & Automation student at ENSEEIHT

SKILLS:

Programming Languages: C, C++, C#, Python, SQL, Ladder Logic (PLC)
Development Tools: Visual Studio, KiCad, LabVIEW, Quartus, MPLAB X IDE, STM32CubeIDE
Communication interfaces: I²C, UART, RS-232, RS-485
Microcontrollers: STM32, Arduino, Microchip
Languages: French (fluent), English (B2)

EDUCATION:

Engineering Degree: Electronics, Electrical Engineering, Energy & Automation **2026-2029**
ENSEEIHT | *Toulouse INP* | *Toulouse* | *France*

Bachelor of Technology in Electronics and Embedded Systems **2024-2026**
IUT Saint-Étienne | *Université Jean Monnet* | *Saint-Étienne* | *France*

Baccalaureate in Mathematics – Highest Honors **2024**
Lycée El Macir | *Algiers* | *Algeria*

PROFESSIONAL EXPERIENCE:

Research & Development Intern **April-June 2026**
NBA Techno | *Saint Jean-Bonnefonds* | *France*

- Analyzed the STM32 firmware of a rehabilitation machine and developed a serious game for patients in Python (Pygame).
- Programmed a PIC18F66K40 microcontroller in C (MPLAB X IDE) and implemented UART communication over an RS-485 bus.
- Soldered PCBs and tested electronic boards, using an oscilloscope and a logic analyzer to check the capacitive sensor signals.

Student Ambassador – Electrical Engineering Department **2026**
- Represented the department of Université Jean Monnet (Saint-Étienne) at the Lyon Student Forum.

First-Year Student Tutor in Mathematics and Electronics **2026**
- Prepared explanations and exercises to help first-year students understand the course material.
- Reviewed and corrected students' exercises.

ACADEMIC PROJECTS:

Line-Following Formula 1 Robot: Designed and programmed a line-following robot using:
- QRE1113 reflective infrared sensors to detect the black line
- an MCP23017 16-bit I/O expander, read over I²C, to acquire the 16 sensor states

Accelerometer Signal Processing: Acquired and processed accelerometer signals in LabVIEW to extract motion information.
Hardware Security – CSAW (ChipWhisperer): Flashed the provided firmware onto the target board and recovered its password character by character with a power-analysis side-channel attack, using a scripted loop to compare power traces.

INTERESTS:

Sports: basketball, badminton
Painting: work shown in several exhibitions
Travel: USA, Canada, Spain, Turkey